

HOTEL OS

Guest Super-App

PRD · Build Plan · Input/Output Specification

Claude Code Ready | Complete & Self-Contained

Version 1.0 · Confidential

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Part 1 — Product Requirements Document (PRD)

1. Executive Summary

Hotel OS is a mobile-first super-app for hotel guests, paired with a staff operations companion app. It replaces fragmented, paper-based, and front-desk-dependent hotel operations with a single elegant digital experience on the guest's phone. The product delivers contactless check-in, mobile room keys, room service ordering, laundry and housekeeping management, a 24/7 AI concierge, loyalty rewards, and smart bill payment — all from one app.

For hotel staff, the companion app provides a real-time task queue, guest intelligence, live occupancy data, and direct communication tools. The web dashboard gives management a full analytics command center.

Target launch: Phase 1 MVP in 10 weeks. Full Phase 2 at week 20. Target: 3 pilot hotels live before wider rollout.

2. Problem Statement

2.1 Guest Pain Points

- Check-in queues waste 10–20 minutes of a guest's arrival experience
- No way to order room service or request housekeeping without calling the front desk
- No visibility into order status — guests wait in uncertainty
- Loyalty programmes are invisible during the stay — guests don't know their points balance
- Feedback is limited to post-checkout forms that few guests fill out
- Lost room keys interrupt stays and consume front-desk time

2.2 Hotel Operations Pain Points

- Staff rely on radio/phone to coordinate tasks — no structured system
- No centralised view of what every guest needs right now
- Housekeeping status tracked on paper clipboards — prone to error
- Revenue leaks from missed upsell moments (upgrade, late checkout, F&B)
- Guest complaints surface on TripAdvisor, not internally — too late to fix
- OTA dependency costs 15–25% commission — no direct booking incentive for guests

3. Goals & Success Metrics

Goal	Success Metric (90 days post-launch)
Reduce check-in time	Average front-desk check-in time < 3 minutes (from ~12 min)
Increase F&B revenue	In-app F&B orders ≥ 25% of total F&B revenue per property
Reduce front-desk calls	Guest service calls reduced by 40% (app self-service replacement)
Loyalty engagement	≥ 60% of guests open the app at least once per stay
Guest satisfaction	In-app feedback score ≥ 4.2 / 5.0 average
Staff task efficiency	SLA compliance (on-time completion) ≥ 85% for all order types
Direct booking uplift	≥ 15% of returning guests book directly through the app (not OTA)

4. Users & Personas

4.1 Primary User — The Hotel Guest

Attribute	Detail
Age range	25–55 years old
Devices	Android (60%) and iOS (40%) — India market skews Android
Tech comfort	Moderate — uses food delivery and payments apps daily
Key need	Convenience and speed. Does not want to call the front desk.
Secondary need	Recognition as a loyal guest. Wants to feel the hotel "knows" them.
Frustration point	Waiting. Uncertainty. Repeating preferences every stay.

4.2 Secondary User — Hotel Staff (Front Desk / Housekeeping / F&B)

Attribute	Detail
Role types	Front desk, housekeeping supervisor, room service, maintenance, manager
Devices	Hotel-issued Android tablets (staff app). Desktop for dashboard.
Tech comfort	Low-to-moderate. UI must be simple, fast, and require zero training.
Key need	See what needs to be done right now, in priority order. No confusion.
Frustration point	Radio calls interrupting tasks. Not knowing if a guest complaint is resolved.

4.3 Tertiary User — Hotel Manager / Owner

Attribute	Detail
Platform	Web dashboard (desktop). Mobile app for alerts on the go.
Key need	Revenue overview, occupancy, guest sentiment, staff performance — at a glance.
Decision context	Uses data to adjust pricing, staffing levels, and marketing spend.

5. Product Scope

5.1 In Scope — Phase 1 (MVP, weeks 1–10)

- Guest app: OTP login, reservation view, AI concierge chat, food & beverage ordering
- Guest app: Digital pre-check-in (ID scan, preferences), mobile room key (BLE/NFC)
- Guest app: Real-time order status tracking, in-app bill view, basic loyalty points display
- Staff app: Task queue, order management (accept/update/complete), push alerts
- Backend: auth-service, booking-service, orders-service, access-service (basic)
- Backend: PostgreSQL schema, Redis cache, Kong gateway, Fastify APIs
- AI: Claude-powered concierge with RAG knowledge base, intent classification
- Integration: Cloudbeds PMS webhook sync, Razorpay payment, Firebase FCM push

5.2 In Scope — Phase 2 (weeks 11–20)

- Guest app: Laundry tracking, housekeeping scheduling, DND toggle, amenity requests
- Guest app: Express checkout with auto-bill summary, group bill split
- Guest app: Full loyalty programme UI — points history, tier progress, challenges
- Guest app: Emotion-aware feedback system (mood picker + voice note option)
- Staff app: Housekeeping floor map, guest profile view, feedback alerts
- Web dashboard: Occupancy overview, revenue metrics, order analytics, staff performance
- Backend: loyalty-service (full), payments-service (invoicing), housekeeping module

5.3 Out of Scope (deferred to Phase 3+)

- AI WhatsApp CRM (separate product — Phase 3)
- IoT room environment control (smart AC, lighting)
- AR local explorer
- Digital memory book (post-stay photo album)
- Multi-property management dashboard
- Voice concierge (Whisper TTS/STT)
- Demand forecasting and dynamic pricing engine

6. Feature Specifications

6.1 Guest App — Authentication

Attribute	Specification
Login method	Mobile phone OTP — 6 digits, expires in 5 minutes, max 3 attempts
OTP provider	Firebase Auth SMS (primary), Twilio fallback
Biometric	Face ID / fingerprint via expo-local-auth after first OTP login
Session	JWT access token (15 min TTL) + refresh token (30 days, rotated)
New guest flow	Enter phone → OTP → name → done. No password, no email required.
Returning guest	Enter phone → OTP → lands on home screen with active reservation.
Privacy	Phone number is the only mandatory PII. Email optional.

6.2 Guest App — Home Screen

The home screen is contextual. Its content changes based on the guest's current state.

State	Home Screen Shows
No reservation	Search hotels CTA, loyalty points, past stays shortcut
Reservation confirmed (pre-arrival)	Countdown to check-in, digital check-in CTA, room preview, weather widget
Checked in (active stay)	Quick action grid (order food, housekeeping, key, concierge), room number, loyalty points, bill balance
Checked out (post-stay)	Rate your stay CTA, bill receipt download, loyalty earned this stay, book again CTA

6.3 Guest App — Digital Check-In

Step	Detail
Trigger	Enabled 24 hours before check-in date. Push notification sent.
ID Scan	Guest scans passport or Aadhaar using expo-camera + Google MLKit OCR. Name and DOB extracted and pre-filled.
Preferences	Pillow type, AC temperature, floor preference (high/low), early check-in request, special notes.
Confirmation	Backend validates ID scan hash. Room pre-assigned. PMS record updated via Cloudbeds API.

Key provisioning	Mobile key generated in access-service on check-in confirmation. Delivered to app via secure API.
Physical key fallback	Front desk can always issue a physical key. App check-in is parallel, not mandatory.

6.4 Guest App — Mobile Room Key

Attribute	Specification
Protocol	Bluetooth Low Energy (BLE) — works offline once key is provisioned
Activation	Key activates on actual_check_in timestamp. Deactivates on checkout.
UI interaction	Hold phone near lock → haptic feedback + visual unlock confirmation
Lock hardware	ASSA ABLOY Visionline (primary), Salto KS (secondary)
Security	Key stored in Expo SecureStore (device secure enclave). Never in app memory or logs.
Sharing	Explicitly out of scope. One key per reservation only.
Lost phone	Manager can remotely revoke key via staff app. Physical key issued as fallback.

6.5 Guest App — Food & Beverage Ordering

Attribute	Specification
Menu display	Category tabs (Breakfast / Mains / Desserts / Beverages / Room Extras). Allergen tags visible. AI recommendations shown at top.
Ordering	Add to cart, adjust quantity, add item-level notes (e.g. "no onions"), set delivery time (ASAP or scheduled).
Payment	Charge to room folio (default). Also: UPI, card, loyalty points redemption.
Order tracking	Live status bar: Received → Confirmed → Preparing → Out for Delivery → Delivered. Socket.io real-time.
Rating	After delivery: 1–5 star tap rating + optional text note. Appears in staff performance metrics.
Reorder	One-tap reorder of previous order (last 3 orders shown on menu home screen).
Kitchen hours	Menu items greyed out with "Available from 7am" message outside kitchen hours.

6.6 Guest App — Housekeeping & Laundry

Feature	Specification
DND toggle	Activates Do Not Disturb. Housekeeping app shows room as DND. Override only by manager.
Make up room	Request room cleaning now. Creates priority housekeeping task.
Schedule housekeeping	Pick time slot (9am–5pm in 1hr blocks). Assigned to staff calendar.
Extra amenities	Request extra towels, toiletries, hangers, adapters etc. from predefined list.
Laundry pickup	Tap to request pickup. Staff notified. Items count entered by guest (5 shirts, 2 trousers, etc.).
Laundry status	Real-time: Pickup Requested → Collected → Washing → Drying → Ready → Delivered.
Laundry turnaround	Same-day if picked up before 10am. Next-day if after 10am. Shown in UI.

6.7 Guest App — AI Concierge

Attribute	Specification
Interface	Chat UI (WhatsApp-style). Text input + voice note option.
Availability	24/7. Sub-3 second response time P95.
Knowledge base	Hotel menu, room types, policies, check-out time, pool/gym hours, parking, local attractions, FAQ — all embedded in Pinecone vector DB per property.
Capabilities	Answer questions, create orders via chat ("order a club sandwich"), report issues, request services, provide local recommendations.
Language	Auto-detects guest language from phone locale. Responds in same language. Supports English, Hindi, Tamil, Telugu, Kannada initially.
Escalation	If confidence < 0.7 or complaint detected: offers to connect with human staff. Human receives full chat context.
Memory	Within-stay context retained. Remembers preferences mentioned in conversation.

6.8 Guest App — Loyalty & Rewards

Feature	Specification
Points display	Current balance always visible on home screen header. Tap opens full loyalty screen.
Earn events	Room stay (1pt/₹100), F&B orders (2pt/₹100), App check-in (+50pt), Review submitted (+100pt)

Tier levels	Bronze (0), Silver (1,000 pts), Gold (5,000 pts), Platinum (15,000 lifetime pts)
Tier benefits	Bronze: basic. Silver: late checkout request. Gold: room upgrade request, priority service. Platinum: all above + complimentary amenities.
Redemption	Redeem points against room bill or F&B orders. Minimum redemption: 500 points = ₹50.
Statement	Scrollable transaction history: earned, redeemed, expired, bonus — with date and description.
Progress bar	Visual bar showing points to next tier. Animated fill.

6.9 Guest App — Bill & Payment

Feature	Specification
Folio view	Itemised bill: room charges, F&B orders, laundry, extras. Timestamps on each line item.
Live balance	Running balance visible on home screen during stay.
Express checkout	Tap to checkout. Summary shown. Pay via UPI/card/loyalty. Digital receipt emailed.
Group split	Add guest phone numbers. Assign specific charges to each person. Each pays their share via their own payment method.
Invoice download	PDF invoice downloadable from app + emailed automatically on checkout.
Partial payment	Pay a partial amount against folio (e.g., pay F&B charges now, room on checkout).

6.10 Guest App — Feedback System

Attribute	Specification
Triggers	Post-order rating (optional, shown 2 min after delivery). End-of-stay rating (shown on checkout screen).
UI	Emoji mood selector (5 levels from very unhappy to delighted) + optional text note + optional voice note (60 sec max).
Routing	Score ≤ 2 : creates internal complaint ticket, alerts duty manager immediately. Score 4–5: prompts Google/TripAdvisor review redirect.
Per-service rating	Separate ratings for F&B, housekeeping, front desk, concierge — aggregated in dashboard.
Anonymity	Guest can submit anonymously. Anonymous feedback still routed by category.

7. Staff App — Feature Specifications

7.1 Staff App — Task Queue

Feature	Specification
View	Sorted list of pending tasks. Default sort: SLA deadline ascending (most urgent first).
Task card	Shows: guest name, room number, task type (icon), description, time received, SLA countdown (red if < 10 min).
Actions	Swipe right: Accept. Swipe left: Reassign. Tap: view details + update status.
Status updates	Accepted → In Progress → Completed. Each tap records timestamp.
Photo completion	For housekeeping tasks: camera photo required on completion. Uploaded to S3.
Filters	By type: All / Food / Housekeeping / Laundry / Maintenance. By status: Pending / Mine / Completed.
Push alerts	New high-priority task: immediate push. SLA breach approaching: 10-min warning push.

7.2 Staff App — Guest Intelligence Panel

Feature	Specification
Access	Manager and front-desk roles only. Search by name or room number.
Shows	Guest name, tier, total stays, current reservation details, dietary flags, room preferences, AI-generated brief.
AI brief	1–2 paragraph summary generated by Claude: "Rajesh is a Gold member on his 6th stay. Prefers high-floor rooms. Vegetarian. Last stay: positive experience with F&B team. Complained about AC noise in Nov 2024 — resolved."
Active orders	All in-progress orders for this guest shown inline.
Feedback history	Last 3 ratings from this guest shown.

7.3 Staff App — Housekeeping Floor Map

Feature	Specification
View	Floor selector (dropdown). Grid of room cards per floor.
Room status colours	Green = Clean. Yellow = Occupied/dirty. Red = DND. Blue = Inspect needed. Grey = Out of order.
Tap action	Tap room → see full status, assigned cleaner, last cleaned time, guest DND preference.

Assignment	Manager can drag-assign rooms to cleaners from this view.
Progress	Floor completion percentage bar at top.

8. Web Dashboard — Feature Specifications

Module	What It Shows
Overview	Live occupancy %, today's arrivals/departures, RevPAR, ADR, revenue vs last week. Refreshes every 60s.
Operations	All pending + in-progress tasks across property. SLA breach count. Housekeeping floor map (same as staff app).
Guest list	Searchable, sortable table of all in-house guests. Filter by tier, sentiment score, checkout date.
Order analytics	Order volume by type, average rating per category, SLA compliance %, peak time chart.
Feedback & NPS	Average rating trends, per-category breakdowns, individual feedback viewer, complaint resolution tracker.
Loyalty	Tier distribution, points issued vs redeemed chart, top guests by loyalty spend.
Staff performance	Task completion rate, average delivery rating, SLA breach count per staff member.
Revenue report	Revenue by segment (room/F&B/laundry/extras). Date range selector. Export to CSV.

9. Non-Functional Requirements

Requirement	Target
API response time (P95)	< 300ms for all read endpoints. < 500ms for write endpoints.
AI concierge response (P95)	< 3 seconds end-to-end (includes RAG retrieval + LLM generation).
Mobile app startup time	< 2 seconds cold start on mid-range Android (Snapdragon 665 class).
Uptime SLA	99.9% per calendar month (< 44 minutes downtime).
Concurrent users	Support 500 concurrent guests per property without degradation.
Offline capability	Mobile key and room number work offline. Order history viewable offline. New orders require connectivity.
Security	JWT RS256, TLS 1.3, AES-256 at rest, OWASP Top 10 hardened, no PII in logs.
Compliance	India DPDP Act 2023 — explicit consent, data deletion within 30 days on request.

Accessibility	WCAG 2.1 AA for web dashboard. VoiceOver / TalkBack support for key app screens.
Localisation	English + Hindi + Tamil + Telugu at launch. String-externalised for easy addition.

Part 2 — User Stories & Acceptance Criteria

Priority Key: P0 = Must-have MVP blocker | P1 = Phase 1 required | P2 = Phase 2 | P3 = Nice-to-have

10. Guest App User Stories

10.1 Authentication

User Story	Priority	Effort (sp)	Phase	Acceptance Criteria
As a guest, I can log in with my phone number and OTP	P0	3	Phase 1	OTP sent in < 5s. Verified OTP returns JWT. Wrong OTP shows error. 3 failures = lockout 15 min.
As a returning guest, my name and preferences are pre-filled after login	P1	2	Phase 1	Guest profile loaded. Home screen shows active reservation if exists.
As a guest, I can use Face ID / fingerprint after initial OTP login	P2	2	Phase 2	Biometric prompt appears on re-open. Falls back to OTP if fails.

10.2 Check-in

User Story	Priority	Effort (sp)	Phase	Acceptance Criteria
As a guest, I can complete digital check-in 24hr before arrival	P0	5	Phase 1	ID scan extracts name+DOB. Preferences saved. Confirmation screen shown. PMS updated.
As a guest, I receive my mobile room key after completing check-in	P0	8	Phase 1	Key active from check-in time. BLE unlock works within 2m of lock. Key revoked on checkout.
As a guest, I can view my room number and arrival details before check-in	P1	2	Phase 1	Reservation screen shows room type, floor, dates, rate plan.
As a guest, I can request an early check-in during digital check-in	P2	3	Phase 2	Request creates staff task. Guest notified when room ready.

10.3 Food & Beverage

User Story	Priority	Effort (sp)	Phase	Acceptance Criteria
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As a guest, I can browse the full F&B menu and place an order	P0	5	Phase 1	Menu loads < 1s. Cart works. Order created in backend. Staff notified.
As a guest, I can track my order status in real time	P0	5	Phase 1	Status updates without refresh. Socket.io. Estimated time shown.
As a guest, I can schedule food delivery for a specific time	P1	3	Phase 1	Time picker (30-min slots). Order held until 10 min before scheduled time.
As a guest, I can rate my food delivery	P1	2	Phase 1	Rating prompt 2 min after Delivered status. 1–5 stars + optional note.
As a guest, I see AI-personalised menu recommendations	P2	5	Phase 2	Top 3 recommendations based on dietary profile + order history shown at menu top.

10.4 Housekeeping & Laundry

User Story	Priority	Effort (sp)	Phase	Acceptance Criteria
As a guest, I can toggle DND to prevent housekeeping entry	P0	2	Phase 1	DND flag set immediately. Housekeeping app shows room as DND. Cannot be overridden by staff (only manager).
As a guest, I can request room cleaning or extra amenities	P1	3	Phase 1	Creates housekeeping task. Staff notified. SLA timer starts. Completion notified.
As a guest, I can schedule housekeeping for a specific time slot	P2	3	Phase 2	Time slot picker. Assigned to staff schedule. Reminder push sent to guest 30 min before.
As a guest, I can request laundry pickup and track its status	P1	5	Phase 1	Pickup request creates task. Status updates through 5 stages. Push notification on ready.

10.5 AI Concierge

User Story	Priority	Effort (sp)	Phase	Acceptance Criteria
As a guest, I can chat with the AI concierge 24/7	P0	8	Phase 1	Chat UI open. Message sent. Response within 3s. Context maintained in session.
As a guest, the concierge can create a room service order from my chat message	P0	5	Phase 1	"Order a club sandwich" → order created → confirmation shown in chat → staff notified.
As a guest, I can ask about hotel amenities and get accurate answers	P1	3	Phase 1	RAG retrieves correct answer from hotel knowledge base. Hallucinations prevented.

As a guest, I can request to speak with a human from the concierge	P1	2	Phase 1	Human handoff button. Staff receives push + full chat transcript. Bot paused for thread.
As a guest, I can chat in my preferred language	P2	5	Phase 2	Language auto-detected. Response in same language. Tamil, Hindi, Telugu, Kannada supported.

10.6 Loyalty

User Story	Priority	Effort (sp)	Phase	Acceptance Criteria
As a guest, I can see my current loyalty points and tier on the home screen	P1	3	Phase 1	Points displayed in header. Tier badge shown. Tap opens loyalty detail screen.
As a guest, I can view my full points transaction history	P2	3	Phase 2	Scrollable list. Earn/redeem/bonus entries with dates and descriptions.
As a guest, I can redeem points against my room bill or F&B orders	P2	5	Phase 2	Redemption UI in checkout and cart. Minimum 500 points. Balance updated instantly.

10.7 Bill & Checkout

User Story	Priority	Effort (sp)	Phase	Acceptance Criteria
As a guest, I can view my running bill during my stay	P1	3	Phase 1	Folio screen. Itemised. Live balance. Timestamp per line item.
As a guest, I can express-checkout and pay via UPI/card	P0	8	Phase 1	Checkout flow: summary → payment → receipt. Razorpay integration. Key revoked on success.
As a guest, I can split the bill with other guests in my group	P2	8	Phase 2	Add guest phone numbers. Assign charges. Each person pays independently.

11. Staff App User Stories

User Story	Priority	Effort (sp)	Phase	Acceptance Criteria
As a staff member, I see all pending tasks in priority order	P0	5	Phase 1	Task list sorted by SLA deadline. Push notification on new task. SLA countdown visible.

As a staff member, I can accept and update the status of a task	P0	3	Phase 1	Status transitions: Pending → Accepted → In Progress → Completed. Timestamps recorded.
As a staff member, I receive push alerts for urgent tasks	P0	3	Phase 1	Push within 30s of task creation. Badge count on app icon updated.
As a manager, I can view all guests and their AI-generated brief	P1	5	Phase 1	Guest list with search. Tap → guest profile with AI summary, tier, active orders.
As a manager, I can see the housekeeping status of all rooms	P2	5	Phase 2	Floor map with colour-coded room status. Assignment controls.
As a manager, I receive an alert when a guest gives negative feedback	P1	3	Phase 1	Score ≤ 2 → instant push to manager + email. Complaint ticket created.

Part 3 — Input / Output Specification

Purpose: This section defines every API endpoint with precise input payloads and output shapes. Use this directly when writing service handlers in Claude Code.

12. API Conventions

Convention	Detail
Base URL	https://api.hotelosapp.io/api/v1
Authentication	Authorization: Bearer {JWT} in header for all protected routes
Content-Type	application/json for all requests and responses
Timestamps	All timestamps in ISO 8601 UTC: 2024-11-15T14:30:00Z
IDs	UUID v4 strings: "550e8400-e29b-41d4-a716-446655440000"
Errors	{ "error": { "code": "VALIDATION_ERROR", "message": "...", "details": [...] } }
Pagination	{ "data": [...], "meta": { "page": 1, "per_page": 20, "total": 143 } }
Empty states	Empty arrays [] not null for list fields

13. Authentication Endpoints

POST /auth/otp/send

Request Body

```
{  "phone": "+919876543210"    // E.164 format, required }
```

Response 200

```
{  "success": true,    "expires_in": 300    // OTP valid for 5 minutes }
```

Response 429

```
{  "error": {    "code": "RATE_LIMITED",    "message": "Too many OTP requests",    "retry_after": 900    // seconds  } }
```

POST /auth/otp/verify

Request Body

```
{  "phone": "+919876543210",  "otp": "847291" }
```

Response 200

```
{  "access_token": "eyJhbGc...",    // JWT, 15 min TTL  "refresh_token": "eyJhbGc...",    // JWT, 30 day TTL  "guest": {    "id": "uuid",    "phone": "+919876543210",    "full_name": "Priya Mehta",    "loyalty_tier": "gold",  }
```

```
"loyalty_points": 4820,      "is_new": false           // true = first login ever
} }
```

14. Guest Profile Endpoints

GET /guests/me

Response 200

```
{  "id": "uuid",    "phone": "+919876543210",    "full_name": "Priya Mehta",
  "email": "priya@email.com",    "language_code": "en",    "loyalty_tier": "gold",
  "loyalty_points": 4820,    "lifetime_points": 12400,    "total_stays": 8,
  "dietary_flags": ["vegetarian"],    "active_reservation": {    "id": "uuid",
  "room_number": "412",    "room_type": "Deluxe King",    "check_in_date": "2024-
11-15",    "check_out_date": "2024-11-17",    "status": "checked_in",
  "balance_due": 4200.00    } }
```

PATCH /guests/me/preferences

Request Body

```
{  "room_temp_celsius": 22,    "pillow_type": "soft",           // soft | firm |
medium    "floor_preference": "high",           // high | low | none    "dietary_flags":
["vegetarian", "gluten-free"],    "newspaper": "Times of India",    "extra_blanket":
true }
```

Response 200

```
{  "success": true,    "updated_fields": ["room_temp_celsius", "pillow_type",
"dietary_flags"] }
```

15. Reservation Endpoints

GET /reservations/active

Response 200

```
{  "id": "uuid",    "pms_booking_ref": "CB-884721",    "status": "checked_in",
  "room": {    "id": "uuid",    "room_number": "412",    "room_type": "Deluxe
King",    "floor": 4,    "amenities": ["pool_view", "balcony"]    },
  "check_in_date": "2024-11-15",    "check_out_date": "2024-11-17",    "adults": 2,
  "children": 0,    "rate_plan": "CP",    "room_rate": 6500.00,    "total_room_amount":
13000.00,    "total_fnb_amount": 1840.00,    "total_other_amount": 0.00,
  "total_amount": 14840.00,    "paid_amount": 6500.00,    "balance_due": 8340.00,
  "mobile_key_status": "active",    "special_requests": "Quiet room please" }
```

POST /reservations/:id/pre-checkin

Request Body

```
{  "id_scan": {    "document_type": "passport",           // passport | aadhaar | dl
  "full_name": "Priya Mehta",    "date_of_birth": "1988-03-22",
  "document_number_hash": "sha256:abc123..." // hashed, never raw    },
```

```
"preferences": {      "room_temp_celsius": 22,      "pillow_type": "soft",
"floor_preference": "high",      "early_checkin_request": false,
"special_notes": "Celebrating anniversary"  },      "eta": "14:00"
// estimated arrival time HH:MM }
```

Response 200

```
{  "success": true,  "checkin_confirmed": true,  "mobile_key_status":
"pending_activation",  // activates at check_in_date  "room_assigned": {
"room_number": "412",      "floor": 4,      "room_type": "Deluxe King"  },
"message": "You're all set! Your mobile key will activate at check-in time." }
```

POST /reservations/:id/checkout

Request Body

```
{  "payment_method": "razorpay",      // razorpay | stripe | loyalty_points | folio
"razorpay_payment_id": "pay...", // from Razorpay SDK callback  "tip_amount":
200.00,      // optional  "bill_split_confirmed": false      // true if
split was pre-arranged }
```

Response 200

```
{  "success": true,  "checkout_time": "2024-11-17T11:23:00Z",  "total_paid":
14840.00,  "loyalty_points_earned": 148,  "loyalty_points_balance": 4968,
"invoice_url": "https://cdn.hotelosapp.io/invoices/INV-2024-004821.pdf",
"invoice_email_sent": true,  "mobile_key_status": "revoked" }
```

16. Orders Endpoints

GET /menu

Query Parameters

```
?category=mains      // breakfast | mains | desserts | beverages | extras
?dietary=vegetarian  // vegetarian | vegan | gluten-free | halal ?search=pasta
// full-text search ?recommended=true      // AI recommendations first
```

Response 200

```
{  "recommended": [  {      "id": "uuid",      "name": "Masala Dosa",
"description": "Crispy crepe with spiced potato filling...",      "price":
280.00,      "category": "breakfast",      "dietary_tags": ["vegetarian",
"gluten-free"],      "allergens": [],      "prep_time_minutes": 15,
"is_available": true,      "image_url": "https://cdn.hotelosapp.io/menu/masala-
dosa.jpg",      "ai_reason": "Based on your vegetarian preference"  }  ],
"items": [...],      // full menu array, same shape  "categories":
["breakfast", "mains", "desserts", "beverages", "extras"],  "kitchen_open": true,
"kitchen_hours": { "open": "07:00", "close": "23:00" } }
```

POST /orders

Request Body

```
{  "reservation_id": "uuid",  "type": "food",      // food | beverage |
laundry | housekeeping | amenity  "items": [  {      "menu_item_id": "uuid",
"name": "Masala Dosa",      "quantity": 2,      "unit_price": 280.00,
```

```
"notes": "Extra chutney please" }, { "menu_item_id": "uuid",
"name": "Filter Coffee", "quantity": 2, "unit_price": 120.00,
"notes": "" } ], "scheduled_for": null, // null = ASAP. ISO
timestamp for scheduled. "payment_method": "folio", // folio | razorpay |
loyalty_points "notes": "Please knock loudly, we're on a call" }
```

Response 201

```
{ "order_id": "uuid", "status": "pending", "total_amount": 800.00,
"items_count": 4, "estimated_delivery_minutes": 25, "sla_deadline": "2024-11-
15T15:55:00Z", "tracking_channel": "socket.io", // listen on
room:{reservation_id}:orders "message": "Order received! Estimated delivery: 25
minutes" }
```

GET /orders/:id — Status Tracking Response

```
{ "id": "uuid", "type": "food", "status": "in_progress", //
pending|accepted|in_progress|completed|cancelled "status_history": [ {
"status": "pending", "at": "2024-11-15T15:30:00Z" }, { "status":
"accepted", "at": "2024-11-15T15:31:30Z" }, { "status": "in_progress",
"at": "2024-11-15T15:34:00Z" } ], "assigned_staff": { "name": "Ravi
Kumar", "role": "room_service" }, "estimated_completion": "2024-11-
15T15:55:00Z", "items": [...], "total_amount": 800.00, "guest_rating": null
// populated after completion }
```

17. Access / Mobile Key Endpoints

GET /reservations/:id/key

Response 200 — Key Active

```
{ "status": "active", "key_token": "ENCRYPTED_BLE_TOKEN", // consumed by
react-native-ble-plx "room_number": "412", "valid_from": "2024-11-
15T00:00:00Z", "valid_until": "2024-11-17T12:00:00Z", "lock_type":
"assa_abloy", "ble_device_id": "LOCK_DEVICE_UUID" }
```

Response 200 — Key Not Yet Active

```
{ "status": "pending_activation", "activates_at": "2024-11-15T00:00:00Z",
"message": "Your key will activate on your check-in date" }
```

18. AI Concierge Endpoints

POST /concierge/message

Request Body

```
{ "reservation_id": "uuid", "message": "Can I get a club sandwich and two
cokes sent to my room?", "message_type": "text", // text | voice_note
"voice_url": null, // S3 URL if message_type is voice_note
"session_id": "uuid" // client-generated, maintains conversation context
}
```

Response 200

```
{  "session_id": "uuid",  "response_text": "Of course! I've placed your order:
1x Club Sandwich (₹420) and 2x Coca Cola (₹160). Total: ₹580 charged to your folio.
Estimated delivery: 20 minutes.",  "actions_taken": [  {  "type":
"order_created",  "order_id": "uuid",  "details": "Club Sandwich + 2x
Coke - ₹580"  }  ],  "intent": "ORDER",  "confidence": 0.97,
"needs_human": false,  "follow_up_suggestions": [  "Track my order",  "Add
a dessert",  "What time is checkout?"  ] }
```

19. Loyalty Endpoints

GET /loyalty/summary

Response 200

```
{  "current_points": 4820,  "lifetime_points": 12400,  "tier": "gold",
"next_tier": "platinum",  "points_to_next_tier": 2600,  "tier_progress_pct":
82.7,  "points_expiring_soon": {  "amount": 500,  "expiry_date": "2025-01-
31"  },  "this_stay_earned": 148,  "redemption_value": 482.00,  // ₹1 per 10
points  "tier_benefits": [  "Late checkout request (subject to availability)",
"Priority room service",  "Room upgrade request"  ] }
```

GET /loyalty/statement

```
{  "data": [  {  "id": "uuid",  "type": "earn",  "points": 148,
"balance_after": 4968,  "reason": "Room stay + F&B - Nov 15-17, 2024",
"reservation_id": "uuid",  "created_at": "2024-11-17T11:23:00Z"  },  {
"id": "uuid",  "type": "redeem",  "points": -500,  "balance_after":
4468,  "reason": "Redeemed against F&B order #ORD-2024-008821",
"created_at": "2024-11-16T19:44:00Z"  }  ],  "meta": { "page": 1, "per_page":
20, "total": 34 } }
```

20. Feedback Endpoints

POST /feedback

Request Body

```
{  "reservation_id": "uuid",  "order_id": null,  // null for stay
rating, order_id for order rating  "rating": 5,  // 1-5 integer
"mood": "delighted",  // very_unhappy | unhappy | neutral | happy |
delighted  "categories": {  "food": 5,  "housekeeping": 4,
"front_desk": 5,  "concierge": 5  },  "text_note": "The room was beautiful
and the staff were amazing!",  "voice_note_url": null,  // S3 URL if voice
recorded  "is_anonymous": false }
```

Response 201

```
{  "feedback_id": "uuid",  "points_earned": 100,  // review bonus
"redirect_to_google": true,  // true if rating >= 4  "google_review_url":
"https://g.page/r/...",  "message": "Thank you! We've credited 100 loyalty points
for your feedback." }
```

21. Staff App Endpoints

GET /staff/tasks

Query Parameters + Response

```
// Query: ?status=pending&type=food&assigned_to=me {  "data": [    {      "id": "uuid",      "type": "food",      "status": "pending",      "priority": "high",      "guest": {        "name": "Priya Mehta",        "room_number": "412",        "loyalty_tier": "gold"      },      "description": "Club Sandwich + 2x Coke",      "total_amount": 580.00,      "created_at": "2024-11-15T15:30:00Z",      "sla_deadline": "2024-11-15T16:00:00Z",      "sla_minutes_remaining": 18,      "is_sla_breached": false    },    {      "summary": {        "total_pending": 4,        "total_sla_breached": 0,        "my_tasks": 2      }    }  ]}
```

PATCH /staff/tasks/:id/status

Request Body + Response

```
// Request: {  "status": "accepted",      // accepted | in_progress | completed | cancelled  "notes": "On my way to room 412",  "completion_photo_url": null // S3 URL, required for housekeeping completion }
// Response 200: {  "success": true,  "task_id": "uuid",  "new_status": "accepted",  "timestamp": "2024-11-15T15:31:30Z",  "guest_notified": true }
// Socket.io update sent to guest app }
```

22. Socket.io Event Reference

Connection: Guests connect to `wss://api.hotelosapp.io` with their JWT. They auto-join `room:{reservation_id}` channel.

Event Name	Payload & When Fired
order:status_update	{ order_id, status, estimated_minutes, staff_name } — fired on every status change
order:delivered	{ order_id, items } — fired when status = completed
key:activated	{ room_number, valid_until } — fired when mobile key becomes active
key:revoked	{ reason } — fired on checkout or manual revoke
task:new (staff)	{ task_id, type, guest_name, room, sla_deadline } — new task created
task:sla_warning (staff)	{ task_id, minutes_remaining } — 10 min before SLA breach
concierge:typing	{ session_id } — AI is generating response (shows typing indicator)
feedback:alert (manager)	{ guest_name, room, rating, category } — rating ≤ 2 triggers this

Part 4 — Complete Build Plan

23. Repository Structure

Monorepo: Use a single monorepo (Turborepo) containing all apps and services. Shared types and utils as internal packages.

```
hotel-os/ | apps/ | mobile/ # React Native guest + staff app
| | web/ # Next.js dashboard | ai-service/ #
Python FastAPI | services/ | auth-service/ | booking-service/ |
| orders-service/ | access-service/ | loyalty-service/ |
payments-service/ | packages/ | types/ # Shared TypeScript
types + Zod schemas | ui/ # Shared React Native UI
components | config/ # ESLint, TS config, Prettier | infra/ |
| terraform/ | k8s/ # Helm charts per service |
docker/ | scripts/ | seed.ts # Database seeder |
migrate.ts | turbo.json | package.json # Root workspace |
docker-compose.yml # Local development stack
```

24. Phase 1 Build Plan — Weeks 1–10

Week 1–2: Foundation

Task	Detail
Monorepo scaffold	pnpm workspaces + Turborepo config. ESLint + Prettier + Husky pre-commit.
PostgreSQL schema	Prisma schema: guests, properties, rooms, reservations, orders, staff tables. Run first migration.
Redis setup	Upstash Redis instance. BullMQ queue config. Test connection from services.
Kong gateway	Docker Compose for local. JWT auth plugin configured. Route definitions for all services.
auth-service	OTP send/verify endpoints. Firebase Auth integration. JWT RS256 issue/verify. Refresh token rotation.
CI pipeline	GitHub Actions: lint → typecheck → test → Docker build → ECR push on main branch.
Dev Docker Compose	All services + PG + Redis runnable locally with single docker-compose up command.

Week 3–4: Booking & PMS Sync

Task	Detail
booking-service	Reservation CRUD. GET /reservations/active. PATCH status. BullMQ event emitters.

Cloudbeds webhook	POST /webhooks/cloudbeds receiver. Parse booking.created, booking.updated, booking.cancelled. Upsert to reservations table.
PMS sync job	BullMQ repeatable job every 5 min to reconcile any missed webhook events via Cloudbeds REST API.
Guest app — auth screens	React Native: splash → phone entry → OTP screen → home. Zustand auth store. JWT stored in MMKV.
Guest app — reservation screen	Reservation detail screen: room info, dates, status, folio balance. Pulls from /reservations/active.

Week 5–6: Orders & Staff App

Task	Detail
orders-service	Full order state machine. POST /orders. PATCH /orders/:id/status. SLA timer via BullMQ delayed job.
Socket.io setup	Socket.io server in orders-service. Guest joins room:{reservation_id}. Staff joins property:{property_id}.
Guest app — menu & cart	Menu screen with categories. Cart state (Zustand). Order placement. POST /orders call.
Guest app — order tracking	Live order status bar consuming Socket.io events. order:status_update → update UI.
Staff app scaffold	Separate entry point in React Native codebase. Role-based routing: staff role → task queue home.
Staff app — task queue	Task list screen. Real-time task updates via Socket.io. Accept/Complete swipe actions. Push alerts via FCM.

Week 7–8: Check-in, Mobile Key & AI

Task	Detail
Digital check-in flow	Guest app: pre-checkin screen. expo-camera + MLKit OCR for ID scan. Preferences form. POST /reservations/:id/pre-checkin.
access-service	ASSA ABLOY Cloud API integration. Key provisioning on booking.checked_in event. GET /reservations/:id/key. Revoke on checkout.
BLE key in app	react-native-ble-plx integration. Unlock flow: retrieve key token → scan for lock BLE ID → connect → write unlock command → haptic feedback.
ai-service scaffold	FastAPI app. Pinecone connection. Embeddings pipeline. Hotel knowledge base ingestion script.
AI concierge — basic	POST /concierge/message. Claude API call with RAG context. Intent classification. Order creation action. Chat UI in guest app.

Week 9–10: Payments, Checkout & QA

Task	Detail
payments-service	Razorpay order creation + payment verification. POST /reservations/:id/checkout. Invoice PDF generation via PDFKit. Email via SendGrid.
Folio screen	Guest app: itemised bill screen. Live balance. Pay now CTA.
Express checkout flow	Guest app: checkout summary → Razorpay payment sheet → success → key revoked → receipt screen.
Loyalty points display	Points balance in home screen header. Basic loyalty summary screen.
Phase 1 QA	Full user flow testing: check-in → order food → track order → checkout. Load test: 100 concurrent guests. Fix all P0/P1 bugs.
Pilot deployment	Deploy Phase 1 to first pilot hotel. Staff training session. Monitor logs + errors for 1 week.

25. Phase 2 Build Plan — Weeks 11–20

Week 11–12: Housekeeping & Laundry

Task	Detail
Housekeeping module	orders-service: housekeeping order type. DND flag on reservations. POST /housekeeping/schedule. PATCH /housekeeping/dnd.
Laundry workflow	5-stage laundry status machine. Pickup → Washing → Drying → Ready → Delivered. Socket.io events per stage.
Guest app — housekeeping screen	DND toggle. Make up room CTA. Schedule time picker. Amenity request selector.
Guest app — laundry screen	Pickup request. Item count form. Live status tracker.
Staff app — housekeeping	Floor map. Room status grid. Assignment controls.

Week 13–14: Full Loyalty Engine

Task	Detail
loyalty-service	Full event-sourced ledger. Earn rule engine (configurable per property). Tier calculation on checkout. Expiry job (nightly cron).
Redemption flow	Guest app cart + checkout: "Use points" toggle. POST /loyalty/redeem. Balance updated instantly.
Loyalty UI	Full loyalty screen: points balance, tier badge, progress bar, challenges grid, transaction history.

Bonus events	App check-in bonus (50 pts), review submission bonus (100 pts), referral bonus (500 pts) — event handlers wired.
---------------------	--

Week 15–16: Feedback & Intelligence

Task	Detail
Feedback system	POST /feedback. Mood picker UI. Voice note recording (expo-av). Sentiment scoring via VADER. Complaint routing.
Manager alerts	Complaint (rating ≤ 2) → Socket.io feedback:alert to manager + FCM push + email.
Guest intelligence	Staff app: guest profile panel. AI brief generation (Claude API call on demand). Active orders + past ratings shown.
AI brief generation	ai-service: POST /ai/guest-brief. Fetches guest record + MongoDB profile + last 3 stays. Returns 2-paragraph brief.

Week 17–18: Bill Split & Express Checkout

Task	Detail
Group bill split	POST /reservations/:id/folio/split. Split definition screen in guest app. Individual Razorpay payment intents per guest.
Partial payments	Pay specific folio line items (e.g. F&B only). Partial payment tracking in reservations table.
Invoice enhancements	Hotel branding on PDF invoice. Itemised breakdown with category subtotals. QR code linking to digital version.

Week 19–20: Web Dashboard & Final QA

Task	Detail
Dashboard scaffold	Next.js 14 app. NextAuth staff login. Protected routes by role.
Overview module	Live occupancy card, today arrivals/departures, revenue today, RevPAR. Auto-refresh every 60s via SWR.
Operations module	Task queue table. Housekeeping floor map (same data as staff app). SLA breach counter.
Analytics modules	Order volume chart (Recharts). Feedback NPS chart. Staff performance table. Revenue by segment chart.
Guest list	Searchable table. Filters: tier, sentiment, checkout date. Tap → guest detail panel.
Phase 2 QA + Pilot 2	Full regression test all P0–P2 features. Second pilot hotel live. Post-launch monitoring.

26. Database Migration Order

Migration	Tables Created
001_initial	properties, staff
002_guests	guests
003_rooms	rooms
004_reservations	reservations
005_orders	orders, order_items
006_loyalty	loyalty_transactions
007_feedback	guest_feedback
008_campaigns	crm_campaigns (schema only, deferred for CRM product)
009_housekeeping	housekeeping_tasks
010_notifications	notification_log

27. Environment Setup for Claude Code

For Claude Code: Copy the block below into CLAUDE.md at the root of the monorepo. This tells Claude Code exactly how the project is structured and how to run it.

```
# Hotel OS - CLAUDE.md ## Project structure Monorepo using pnpm workspaces +
Turborepo. - apps/mobile      React Native 0.74 + Expo SDK 51 (guest + staff app) -
apps/web      Next.js 14 App Router (management dashboard) - apps/ai-service
Python 3.12 + FastAPI (AI/ML service) - services/*      Node.js 20 + Fastify
microservices - packages/types  Shared Zod schemas + TypeScript types ## Commands
pnpm dev      Start all services locally (docker-compose + Turborepo) pnpm
build         Build all packages pnpm test              Run Vitest (Node) + Pytest
(Python) pnpm migrate      Run Prisma migrations pnpm seed      Seed demo
property + guest data ## Key conventions - All API handlers validate input with
Zod FIRST, then process - Never use any type. Use strict TypeScript everywhere. -
All service handlers follow: validate -> auth check -> business logic -> emit event -
Database queries via Prisma only. No raw SQL except for analytics. - BullMQ for
all async work. Never block the request thread with slow operations. - Environment
variables from .env.local (never committed). See .env.example. ## Testing - Unit:
Vitest. Every service function has a unit test. - Integration: Supertest against
real DB (test containers). - API tests use test user with role = guest seeded in
migration 010.
```

28. Key Third-Party Account Setup Checklist

Service	Action Required	Priority	Where Used
Firebase	Create project. Enable Phone Auth. Download service account JSON.	P0 (Week 1)	OTP login, FCM push

Razorpay	Create account. Generate live key_id + key_secret. Enable UPI.	P0 (Week 9)	Payments, checkout
Anthropic	Get API key. Request Claude 3.5 Sonnet access.	P0 (Week 7)	AI concierge
Pinecone	Create account. Create index: hotel-knowledge-base (1536 dims, cosine).	P0 (Week 7)	RAG vector store
Meta WhatsApp	NOT needed for Hotel OS Phase 1. Deferred to CRM product.	Deferred	CRM only
ASSA ABLOY	Contact ASSA ABLOY Cloud partner for API credentials and test lock device.	P1 (Week 7)	Mobile room key
Cloudbeds	Get partner API access. Configure webhook URL after deploy.	P0 (Week 3)	PMS sync
SendGrid	Create account. Verify sender domain. Create invoice + checkout email templates.	P1 (Week 9)	Invoice emails
AWS	Create account. Set up ECR, RDS Aurora, S3 bucket, Secrets Manager.	P0 (Week 1)	All infrastructure
OpenAI	Get API key for text-embedding-3-small (embeddings only).	P0 (Week 7)	RAG embeddings

29. Definition of Done — Per Phase

Phase 1 is done when:

- A new guest can log in, view their reservation, and complete digital pre-check-in
- A guest can receive and use a mobile BLE room key to open their room
- A guest can browse the full menu, place a food order, and track it in real time
- A guest can chat with the AI concierge and place an order through conversation
- A guest can view their folio, complete express checkout, and receive a PDF invoice
- Staff can see all tasks in the staff app, accept them, and mark complete
- The system handles 100 concurrent guests without API response time exceeding 500ms
- All P0 user stories pass their acceptance criteria
- Zero critical (P0) bugs in production after 48 hours live at pilot hotel

Phase 2 is done when:

- Guests can manage laundry through all 5 status stages

- Guests can toggle DND and schedule housekeeping
- Full loyalty programme is live — earn, tier, progress, redeem
- Emotion-aware feedback system is live with manager complaint alerts
- Group bill split works end-to-end with independent payments
- Web dashboard shows live occupancy, order queue, and guest list
- All P1 and P2 user stories pass their acceptance criteria
- Second pilot hotel is live with positive staff adoption (> 70% of orders via app)

Appendix — Quick Reference

A. Priority Definitions

Priority	Definition
P0 — Must Have	Blocks MVP launch. Product cannot go live without this. Build in Phase 1, Week 1–10.
P1 — Should Have	Required for a complete, shippable product. Build in Phase 1 but not a launch blocker.
P2 — Nice to Have	Meaningfully improves the product. Build in Phase 2 (Week 11–20).
P3 — Deferred	Good idea but not yet scoped. Add to backlog for future sprints.

B. Effort Estimation (Story Points)

Story Points	Effort Level
1–2 points	Simple CRUD endpoint or minor UI change. 1–4 hours.
3–5 points	Medium complexity — new screen + API + state management. Half day to 2 days.
8 points	High complexity — third-party integration, complex state machine, or AI feature. 3–5 days.
13 points	Very high — do not plan. Split into smaller stories before adding to sprint.

C. Error Code Reference

Error Code	Meaning
AUTH_REQUIRED	No JWT or expired JWT. Guest must re-authenticate.
FORBIDDEN	JWT valid but role does not have permission for this action.
NOT_FOUND	Resource (reservation, order, etc.) not found or not accessible to this user.
VALIDATION_ERROR	Request body failed Zod schema validation. details[] contains field-level errors.
RATE_LIMITED	Too many requests. retry_after field in error body.
OTP_EXPIRED	OTP has expired (> 5 minutes old). Request a new one.
OTP_MAX_ATTEMPTS	3 failed OTP attempts. Account locked for 15 minutes.
RESERVATION_NOT_ACTIVE	Guest has no active/upcoming reservation. Many features require an active stay.

KITCHEN_CLOSED	F&B order attempted outside kitchen operating hours.
KEY_NOT_ACTIVE	Mobile key accessed before check-in date or after checkout.
PAYMENT_FAILED	Razorpay/Stripe payment verification failed. Show retry UI.
LOYALTY_INSUFFICIENT	Guest tried to redeem more points than their balance.
CONCIERGE_UNAVAILABLE	ai-service unavailable. Show "Try again shortly" and offer human escalation.

D. Data Model Relationships

```
properties 1—< rooms properties 1—< staff properties 1—< guests (via
property_id) guests      1—< reservations rooms      1—< reservations
reservations 1—< orders guests      1—< loyalty_transactions reservations 1—<
guest_feedback crm_campaigns 1—< crm_campaign_sends (M:M with guests)
```

Hotel OS

PRD · Build Plan · Input/Output Specification

Version 1.0 · Ready for Claude Code